

Comprehensive Notes on Statistics: Data Types, Central Tendency, Dispersion, and Correlation

Statistics is the science of collecting, analyzing, presenting, and interpreting data. To effectively analyze data, one must understand its foundational structures, starting from the types of data to advanced relational metrics like correlation.

1. Types of Data

Data can be broadly classified into two primary categories: Qualitative (Categorical) and Quantitative (Numerical). These are further broken down into four distinct levels of measurement.

Qualitative (Categorical) Data

Qualitative data represents characteristics or attributes that cannot be measured numerically. It describes qualities or categories.

- **Nominal Data:** Data split into distinct categories that have no inherent mathematical order or ranking. Examples include gender, eye color, or country of residence.
- **Ordinal Data:** Data categorized into groups that possess a logical order or ranking, but the mathematical distance between the categories cannot be quantified. Examples include satisfaction ratings (Poor, Fair, Good, Excellent) or economic status (Low, Middle, High).

Quantitative (Numerical) Data

Quantitative data represents measurable quantities expressed as numbers. It is categorized based on whether it is countable or continuous.

- **Discrete Data:** Countable values that can only take specific, distinct inputs (usually integers). Examples include the number of children in a family or the number of cars in a parking lot.
- **Continuous Data:** Measurable values that can take an infinite number of possibilities within a given range. Examples include height, weight, and temperature. Continuous data is subdivided into:
 - **Interval Data:** Ordered numeric data where the difference between values is meaningful, but there is no true or absolute zero point. A value of zero does not

mean the absence of the property. Example: Temperature in Celsius or Fahrenheit (0 degrees doesn't mean there is no heat).

- **Ratio Data:** Ordered numeric data with meaningful differences and a strict, absolute zero point, where zero signifies the complete absence of the property. Example: Weight, height, or income (0 dollars means no income).

Data Level	Type	Has Ordered Structure?	Meaningful Differences ?	True Absolute Zero?	Example
Nominal	Qualitative	No	No	No	Marital Status, Blood Type
Ordinal	Qualitative	Yes	No	No	Exam Grades (A, B, C), Survey Likert Scales
Interval	Quantitative	Yes	Yes	No	IQ Scores, Calendar Years
Ratio	Quantitative	Yes	Yes	Yes	Distance, Salary, Age

2. Measures of Central Tendency

Measures of central tendency identify a single central value that represents the center point or typical value of a dataset.

Mean (Arithmetic Average)

The mean is calculated by summing all observations and dividing by the total number of data points. It is highly sensitive to extreme values (outliers).

Population Mean (μ) = $\Sigma X / N$

Sample Mean ($\bar{x}$) = $\Sigma x / n$

Where Σ represents the sum, X/x represents individual values, and N/n represents the total number of points.

Median

The median is the middle value when a dataset is arranged in ascending or descending order. It splits the data exactly in half and is robust against outliers.

- If the number of data points (n) is **odd**: The median is the value at position $(n + 1) / 2$.
- If the number of data points (n) is **even**: The median is the average of the two middle values at positions $(n/2)$ and $(n/2 + 1)$.

Mode

The mode is the value that appears most frequently in a dataset. A dataset can have one mode (unimodal), two modes (bimodal), multiple modes (multimodal), or no mode at all if every value occurs with equal frequency. The mode is the only measure of central tendency applicable to nominal data.

3. Measures of Dispersion (Variability)

Measures of dispersion quantify how spread out or scattered the data points are around the central value.

Variance

Variance measures the average squared distance of each data point from the mean. Squaring the differences ensures negative differences do not cancel out positive ones, but it shifts the metric into squared units.

Population Variance (σ^2) = $\Sigma (X - \mu)^2 / N$

Sample Variance (s^2) = $\Sigma (x - \bar{x})^2 / (n - 1)$

Note: The sample variance uses $(n - 1)$ in the denominator (Bessel's correction) to provide an unbiased estimate of the population variance.

Standard Deviation

The standard deviation is the square root of the variance. It returns the spread metric back to the original units of the data, making it far easier to interpret.

Population Standard Deviation (σ) = $\sqrt{[\sum (X - \mu)^2 / N]}$

Sample Standard Deviation (s) = $\sqrt{[\sum (x - \bar{x})^2 / (n - 1)]}$

A low standard deviation indicates that data points tend to be close to the mean, while a high standard deviation indicates that data points are spread out over a wider range.

4. Percentiles and Quartiles

Percentiles and quartiles are measures of relative standing that indicate the position of a data value relative to the rest of the dataset.

Percentiles

The P-th percentile is the value below which a certain percentage (P%) of the observations fall. For instance, if a score is at the 85th percentile, it means 85% of the values in the dataset are lower than or equal to that score.

Quartiles

Quartiles are specific percentiles that divide an ordered dataset into four equal parts:

- **First Quartile (Q1):** The 25th percentile. It separates the bottom 25% of data from the top 75%.
- **Second Quartile (Q2):** The 50th percentile, which is exactly equivalent to the **Median**.
- **Third Quartile (Q3):** The 75th percentile. It separates the bottom 75% of data from the top 25%.

The **Interquartile Range (IQR)** measures the spread of the middle 50% of the data and is computed as:

IQR = Q3 - Q1. IQR is commonly used to identify outliers via the $1.5 \times \text{IQR}$ rule.

5. Correlations

Correlation evaluates the statistical relationship between two variables, indicating how much they change together.

Pearson Product-Moment Correlation Coefficient (r)

Pearson's r measures the strength and direction of a linear relationship between two

continuous, quantitative variables. The value ranges strictly between -1 and +1.

$$r = \frac{\sum ((x - \bar{x})(y - \bar{y}))}{\sqrt{[\sum (x - \bar{x})^2 * \sum (y - \bar{y})^2]}}$$

Spearman Rank Correlation Coefficient (ρ or r_s)

Spearman's rank correlation evaluates monotonic relationships (whether linear or non-linear) by assessing the ranked values of the variables. It is ideal for ordinal data or non-linear continuous relationships.

Interpreting Correlation Coefficients

- **$r = +1$:** Perfect positive correlation. As variable X increases, variable Y increases proportionally.
- **$r = 0$:** No linear correlation. Changes in X do not predict changes in Y.
- **$r = -1$:** Perfect negative correlation. As variable X increases, variable Y decreases proportionally.

Critical Caveat: Correlation Does Not Imply Causation. Just because two variables move together does not mean one causes the other. A strong correlation can be driven by a third hidden factor (confounding variable) or may simply be a coincidence.